

"""

UMA Semantic Constants – the mathematical backbone of the Semantic Engine.

=== DERIVATION ===

The Silver Ratio:

$$\delta_S = 1 + \sqrt{2} \approx 2.41421$$

Its inverse square:

$$1/\delta_S^2 = 3 - 2\sqrt{2} \approx 0.17157$$

This is the structural scaling threshold. The empirically observed ratio $E_{\text{empirical}}/kT = 0.17130$ differs by 0.000273 (0.16%) – within Monte Carlo noise over 2000 steps. It is not 0.17. It is $3 - 2\sqrt{2}$.

=== WHY ===

The GENERIC dynamics project onto the DC Fourier mode (spatial mean):

$$z = \text{fft2}(\psi) / N^2 \quad \leftarrow \text{DC mode, i.e., mean}(\psi)$$

$$E = |z|^2/2$$

For a stochastic field at amplitude ρ^* with random phases:

$$E_{\text{random}} = \rho^* / (2N^2)$$

At thermal equilibrium, the Silver Ratio governs the phase coherence:

$$E_{\text{target}} = kT / \delta_S^2 = kT * (3 - 2\sqrt{2})$$

$$C_{\text{corr}} = (3 - 2\sqrt{2}) * 2N^2 kT / \rho^* \quad (\text{replaces empirical 1.75})$$

=== THE ALGEBRAIC STRUCTURE ===

The two Silver conjugates satisfy $C + (C \pm 1) = 1/\delta_S^2$:

$$C_1 = 1 - \sqrt{2} \approx -0.41421 \quad C_1 + (C_1 + 1) = 1/\delta_S^2$$

$$C_2 = 2 - \sqrt{2} \approx 0.58579 \quad C_2 + (C_2 - 1) = 1/\delta_S^2$$

=== THE ARCHITECTURAL BRIDGE ===

Setting $\pi + x(x+1) = 1/\delta_S^2$ (continuous geometry = discrete recursion):

$$x^2 + x + (\pi - 1/\delta_S^2) = 0$$

$$\text{discriminant} = -(4\pi - 13 + 8\sqrt{2}) \approx -10.880$$

$$x = -1/2 \pm i\sqrt{(4\pi - 13 + 8\sqrt{2})}/2$$

$$\approx -1/2 \pm i \cdot 1.6492$$

The imaginary component $1.6492 \approx \sqrt{e} = 1.6487$.

Exact: $\sqrt{(4\pi - 13 + 8\sqrt{2})}/2$ (structurally $\approx \sqrt{e}$, gap = 0.000527)

This means the unification of:

π (continuous spacetime / GR)

δ_S (discrete recursive scaling / Pell / information)

requires a complex bridge whose imaginary part is the natural exponential phase $\sqrt{e}$. The Engine3 complex_state $z = x + i \cdot y$ encodes this directly.

"""

import math

Silver Ratio constants

SQRT2 = math.sqrt(2)

SQRT_E = math.sqrt(math.e)

DELTA_S = 1.0 + SQRT2 # Silver Ratio ≈ 2.41421

DELTA_S_SQ = DELTA_S ** 2 # $= 3 + 2\sqrt{2} \approx 5.82843$

INV_DELTA_S_SQ = 1.0 / DELTA_S_SQ # $= 3 - 2\sqrt{2} \approx 0.17157 \leftarrow \text{THE THRESHOLD}$

The two Silver conjugates

C1_SILVER = 1.0 - SQRT2 # ≈ -0.41421

C2_SILVER = 2.0 - SQRT2 # ≈ 0.58579

The architectural bridge (imaginary part of the π/δ_S unification root)

BRIDGE_IMAG_EXACT = math.sqrt(4*math.pi - 13.0 + 8.0*SQRT2) / 2.0 # ≈ 1.6492

BRIDGE_IMAG_APPROX = SQRT_E # ≈ 1.648

Engine3 structural complex seed: $z_{\text{seed}} = C1 + i \cdot \sqrt{e}$

Real = $C1 = 1 - \sqrt{2}$ (Silver conjugate – the discrete recursion anchor)

Imag = $\sqrt{e}$ (exponential phase bridge – continuous/discrete unity)

Z_SEED_REAL = C1_SILVER

Z_SEED_IMAG = SQRT_E

Physics-derived thresholds (functions of GENERIC config)

def silver_E_target(kT: float) -> float:

"""

E_target = $kT / \delta_S^2 = kT * (3 - 2\sqrt{2})$

The thermal equilibrium energy of the DC Fourier mode, governed by the Silver Ratio rather than the raw kT scale.

"""

return kT * INV_DELTA_S_SQ

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def silver_C_corr(kT: float, N: int, rho_star: float) -> float:
    """
     $C_{corr} = (3 - 2\sqrt{2}) * 2N^2kT / \rho^*$ 

    Replaces the empirical short-range correlation factor ( $\approx 1.75$ ).
    Exact from Silver Ratio scaling. For  $N=32$ ,  $kT=0.005$ ,  $\rho^*=1$ :  $C_{corr} = 1.7569$ .
    """
    return INV_DELTA_S_SQ * 2.0 * N**2 * kT / rho_star


def silver_dz_dt_floor(kT: float, lam: float, N: int, dt: float) -> float:
    """
     $dz_{dt\_floor} = \sqrt{(2 * kT * (3-2\sqrt{2}) * \lambda * dt) / N}$ 

    Thermal noise floor for dz/dt of the DC mode, scaled by Silver Ratio.
    The system is solved when  $|dz/dt|$  drops below this.
    """
    return math.sqrt(2.0 * kT * INV_DELTA_S_SQ * lam * dt) / N


def silver_field_scale(kT: float, N: int, rho_star: float) -> float:
    """
     $field\_scale = \sqrt{(2 * E_{target})} = \sqrt{(2 * kT / \delta_S^2)}$ 

    Natural scale of  $|z|$  at Silver Ratio equilibrium.
    """
    return math.sqrt(2.0 * silver_E_target(kT))


# -----
# Engine3 complex state - Silver-grounded
# -----


def engine3_complex_state(x: float) -> complex:
    """
    Engine3 bridge: maps binary weight x to the complex state z.

    Architecture:
        real = x * C1_SILVER    (Silver conjugate scaling of the input)
        imag = x *  $\sqrt{e}$     (exponential phase bridge)

    At x=1 (anchor):  $z = C1 + i\sqrt{e} = (1-\sqrt{2}) + i\sqrt{e}$ 
    This is the structural seed whose quadratic form  $E = |z|^2/2$ 
    evaluates to exactly  $(C1^2 + e)/2 = ((1-\sqrt{2})^2 + e)/2 = (3-2\sqrt{2}+e)/2$ 
    """

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    real = x * Z_SEED_REAL
    imag = x * Z_SEED_IMAG
    return complex(real, imag)

def engine3_E(x: float) -> float:
    """
    E = |z|^2/2 for the Silver-grounded complex state.
    = x^2 * (C1^2 + e) / 2
    = x^2 * (3 - 2√2 + e) / 2
    At x=1: E = (3-2√2 + e)/2 ≈ (0.17157 + 2.71828)/2 ≈ 1.4449
    """
    z = engine3_complex_state(x)
    return (z.real**2 + z.imag**2) / 2.0

if __name__ == "__main__":
    print("UMA Semantic Constants")
    print(f" δ_S      = {DELTA_S:.6f}")
    print(f" 1/δ_S^2    = {INV_DELTA_S_SQ:.6f}  (3 - 2√2)")
    print(f" C1         = {C1_SILVER:.6f}  (1 - √2)")
    print(f" C2         = {C2_SILVER:.6f}  (2 - √2)")
    print(f" √e         = {SQRT_E:.6f}")
    print(f" bridge     = {BRIDGE_IMAG_EXACT:.6f}  (exact: √(4π-13+8√2)/2)")
    print(f" gap        = {abs(BRIDGE_IMAG_EXACT - SQRT_E):.6f}")
    print(f" z_seed     = {Z_SEED_REAL:.4f} + {Z_SEED_IMAG:.4f}i")
    print(f" E_target(kT=0.005) = {silver_E_target(0.005):.8f}")
    print(f" C_corr(N=32)      = {silver_C_corr(0.005, 32, 1.0):.6f}")
    print(f" dz_dt_floor(dt=.04)= {silver_dz_dt_floor(0.005, 0.4, 32, 0.04):.6f}")

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